

Sampling, de Branges, and the windowed Weil form

Where positivity lives; what the contrapositive does and does not say

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1 September 2026

Abstract

Windowed Weil positivity is a sampling inequality for the zeros of ζ in the Paley–Wiener space $PW_{L/2}$. Under RH this is a theorem (*sampling-floor*, Theorem 1: Beurling through the explicit formula). Unconditionally the same quadratic form is Weil’s pairing, not a Gram of real samples: the contrapositive of *global* positivity is RH, but positivity on a single window W_L does not force every zero onto the line. An off-line zero produces a negative direction once the window is large enough to see it; below that scale the form can stay positive. The route does not prove RH. It says, in the language of Beurling density and de Branges spaces, where the positivity of Q_L lives, and which tools of that theory can enter.

1 The sampling reading, already a theorem

Fix $L > 0$, $\tau = L/2$. Even $f \in L^2([-L/2, L/2])$ have Fourier transforms $F \in PW_\tau$. Under RH the explicit formula on this window is

$$Q_L(f) = \sum_{\gamma} F(\gamma)^2,$$

so

$$c_L = \inf_{f \neq 0} \frac{Q_L(f)}{\|f\|^2}$$

is 2π times the lower sampling constant of $\Gamma = \{\pm\gamma\}$ for PW_τ . Beurling–Landau: a separated subset $\Lambda \subset \Gamma$ of lower uniform density $D^-(\Lambda) > \tau/\pi$ is a sampling set. The zeros have $N(T+1) - N(T) \rightarrow \infty$, so such a Λ exists for every τ . Hence $c_L > 0$ and $\lambda_{\min}(V_N) \downarrow c_L$ (*sampling-floor*, Theorem 1). Nothing further is claimed here.

The size of c_L is not a density statement. Density only gives $c_L > 0$. The desert $(-\gamma_1, \gamma_1)$ and the gaps longer than the Nyquist step $2\pi/L$ fix the *exponential* smallness (*sampling-floor* Prop. 2, geometric law of *depth-geometry-quorum-mechanism*). Beurling’s $D^- >$ is too coarse to see $e^{-L\gamma_1}$.

2 Unconditional identity: Weil, not a real Gram

Without RH the left side Q_L is still the prime-side pairing (pole + archimedean – towers $n \leq e^L$). The right side is

$$Q_L(f) = \sum_{\rho} \hat{g}_f(\rho),$$

sum over *all* nontrivial zeros ρ , with $\hat{g}_f$ the entire function attached to f by the explicit formula (Weil–Bombieri normalization of the repository, App. B of *quorum-theorem*). On the line, $\rho = \frac{1}{2} + i\gamma$ and $\hat{g}_f(\rho) = F(\gamma)^2$ (even convention of the window). Off the line, $\rho = \frac{1}{2} + \sigma + i\gamma$ with $\sigma \neq 0$, the term is not a square of a real sample. The form is no longer a Gram of evaluators on $\mathbb{R}$.

Remark. Global Weil positivity — $Q(h) \geq 0$ for every admissible h on $\mathbb{R}$, not just on one W_L — is equivalent to RH. That is Weil’s theorem. Restricting the test class to W_L loses the equivalence: a form that is positive on every compact window is a stronger statement than positivity on one window, and is again RH.

3 Visibility of an off-line zero

Let $\rho = \frac{1}{2} + \sigma + i\gamma$ with $\sigma \neq 0$. The contribution of the pair $(\rho, 1 - \rho)$ to Q_L is $2\operatorname{Re} \hat{g}_f(\rho)$. A test function whose spectral mass sits at ρ can make this term dominate and negative.

Lemma 1 (visibility, schematic). *If $F \in PW_\tau$ can be chosen with $|F(\sigma + i\gamma)| \geq c\|F\|$ and $\sum_\gamma |F(\gamma)|^2$ (the sum over the real zeros) small compared with that, then Q_L takes a negative value as soon as the off-line term is included. Such an F exists in PW_τ once τ exceeds a constant times $|\sigma| + 1$ and the distance from γ to the real sampling nodes is not large compared with τ^{-1} .*

This is Paley–Wiener, not arithmetic: an entire function of type τ cannot peak far off the real axis unless τ is large enough (Phragmén–Lindelöf / Bernstein). Consequence for the repository: positivity of Q_L at $\mu = e^L = 11$ is compatible with an off-line zero at height 10^6 . It is not compatible with an off-line zero inside the band of V_N once N resolves that height. A single certified window does not force RH; a uniform statement over $L \rightarrow \infty$ would.

Wrong continuation. Continuing the Dirichlet-kernel formula $2\sin(sL/2)/(s\sqrt{L})$ to $s = \sigma + i\gamma$ explodes as $\sinh(\gamma L/2)$ and produces $\lambda_{\min} \sim -10^7$ already at $\sigma = 0.05$. That is the Fourier transform in a complex *frequency*, not the Weil pairing. Discarded.

Weil weight. The pairing at $\rho = \frac{1}{2} + \sigma + i\gamma$ is $\hat{\eta}(\rho) = \int_{-L/2}^{L/2} \eta(x) e^{\sigma x} e^{i\gamma x} dx$, bounded by $e^{|\sigma|\tau} \sqrt{L}$. At $\sigma = 0$ it recovers the on-line formula. The quadratic term attached to the pair $(\rho, 1 - \rho)$ is the Gram of $(\hat{\eta}_n(\rho)\hat{\eta}_m(1 - \rho))$. On V_9 at $\mu = 11$, forty real zeros already in Q_{line} :

σ at γ_1	$\lambda_{\min}$ of $Q_{\text{line}} + F(\rho)F(1 - \rho)$	of $Q_{\text{line}} + F(\rho) ^2$
0	$+3 \times 10^{-22}$	$+3 \times 10^{-22}$
0.01	-5.4×10^{-5}	$+3 \times 10^{-17}$
0.10	-7.3×10^{-3}	$+4 \times 10^{-17}$
0.50	-0.23	$+4 \times 10^{-17}$
1.00	-1.28	$+3 \times 10^{-17}$

$|F(\rho)|^2$ never crosses. $F(\rho)F(1 - \rho)$ crosses at once, at height γ_1 , with size $O(\sigma^2)$ at small σ — not 10^7 . The same extra term at height 80 or 200 leaves $\lambda_{\min}$ at 10^{-22} : this window does not see a high off-line zero. Visibility is local in height, mild in σ , and tied to γ_1 . A certified $Q_{11} \succ 0$ still allows an off-line zero at large height. It does not allow one at γ_1 once the pairing is identified with App. B.

App. B closes the “if”. *quorum-theorem*, Remark (i) and App. B: the implemented form is Weil’s pairing for $g = f \star \tilde{f}$, with $\hat{g} = |\hat{f}|^2$ on the critical line, and

$$Q(f) = \sum_{\rho} \hat{g}(\rho)$$

unconditionally. On the line $\hat{g}(\frac{1}{2} + i\gamma) = |\hat{f}(\gamma)|^2$. Off the line $\hat{g}(\rho) = \hat{f}(\rho)\hat{f}(1 - \rho)$. That is the term measured above. The identification is the one already derived for the repository, not a new convention. The factor 2 in $Q = \sum 2\hat{c}\hat{c}^\top$ is the pair $\pm\gamma$.

4 What de Branges supplies, and what it does not

A de Branges space $\mathcal{H}(E)$ is a Hilbert space of entire functions with a reproducing kernel built from a Hermite–Biehler function E . Paley–Wiener PW_τ is the model case $E(z) = e^{-i\tau z}$. Sampling and interpolation in $\mathcal{H}(E)$ are completely described (de Branges, Beurling–Malliavin for the PW case): a set is sampling when its density exceeds the type, with a structure condition on the gaps.

Two precise contacts with Q_L .

PW as $\mathcal{H}(E)$. Theorem 1 is the PW-special case of “the zeros sample $\mathcal{H}(E)$ ”. The desert bound is the reproducing-kernel mass of the hole $(-\gamma_1, \gamma_1)$ in that space — Slepian is the first eigenfunction of the projection onto the hole. Sub-Nyquist gaps are the same statement for a union of holes. The geometric law is a conjecture about the bottom eigenvalue of a Gram of reproducing kernels at the zeros. de Branges gives the language; it does not give $a = 1.69$, $b = 0.82$.

The de Branges approach to ξ . There is a different E , built from ξ , whose space $\mathcal{H}(E)$ has the zeros of ξ as a distinguished set. RH is equivalent to axioms on that E (de Branges 1986 and later variants). That E is *not* $e^{-i\tau z}$. Importing a theorem about PW_τ into the ξ -space is the usual gap in the programme. The windowed form Q_L lives in PW. The ξ -space lives on the critical strip. They share a philosophy (positivity $\Leftrightarrow$ sampling on a line) and not a kernel.

5 What this route can still do in the repository

1. State Theorem 1 as the definition of the floor, not as a numerical finding. Already done in *sampling-floor*.
2. Replace “hyper-nullity” language by “the sampling constant of a set with a hole of width γ_1 and type τ ”. Same numbers, right theory.
3. Prove a quantitative visibility lemma: an explicit $L(\sigma, \gamma)$ beyond which an off-line pair at $\frac{1}{2} + \sigma + i\gamma$ forces $\lambda_{\min}(Q_L) < 0$. This is analysis, not flint.
4. Done on V_9 at $\mu = 11$: the pair at γ_1 crosses; height 80 does not. A quantitative $L(\sigma, \gamma)$ for every height remains analysis.
5. Do *not* claim that $c_L > 0$ for all L unconditionally. That statement is RH.

Status

Theorem 1 (under RH) is already in the living corpus. The present note adds the unconditional reading, the Weil-weight visibility measurement, the identification $\hat{g}(\rho) = \hat{f}(\rho)\hat{f}(1 - \rho)$ with App. B, and the boundary between PW-de Branges and ξ -de Branges. No new theorem. No movement of RH. The branch can stop here.